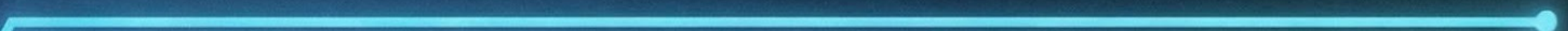




3

OPERATING SYSTEMS



Windows + Active Directory

D3 — Operating Systems



KERBEROS — AD AUTHENTICATION

- Default authentication protocol in Active Directory domains
- KDC (Key Distribution Center) runs on every Domain Controller
- Issues a TGT (Ticket Granting Ticket) after first login — used to request service tickets
- Enables SSO: one login grants access to all authorised domain resources

EXAM SHORTCUT — Kerberos = ticket-based SSO. KDC on DC issues TGT, TGT gets service tickets

EVENT VIEWER — WHICH LOG?

- Security Log → logon events, audit policy hits, object access, privilege use
- Application Log → application errors and crashes
- System Log → OS events, driver failures, hardware errors
- Rule: anything security/audit-related = Security Log

EXAM SHORTCUT — Audit policies and logon events = Security Log. Not Application, not System

WINDOWS REGISTRY — HKLM vs HKCU

- HKEY_LOCAL_MACHINE (HKLM) = system-wide: hardware, drivers, software for ALL users
- HKEY_CURRENT_USER (HKCU) = settings for the currently logged-in user only
- HKLM persists across user sessions; HKCU is per-user

EXAM SHORTCUT — HKLM = machine-wide (all users). HKCU = current user only

Windows + Active Directory

D3 — Operating Systems



GPO PROCESSING ORDER — LSDOU

- Local → Site → Domain → Organizational Unit
- Each layer overrides the previous one — OU wins over Domain wins over Site wins over Local
- GPO applied last = highest priority. OU policy beats everything.
- WSUS client-side targeting: GPO pushes registry keys directing endpoints to update groups

EXAM SHORTCUT — LSDOU: Local, Site, Domain, OU. Last applied = highest priority

DEFAULT ACCOUNT HARDENING

- Rename or disable 'Administrator' and 'root' on all systems
- Default names are known — attacker only needs to guess the password
- Renaming forces attackers to guess BOTH username AND password
- Doubles brute-force complexity; disrupts automated credential attacks

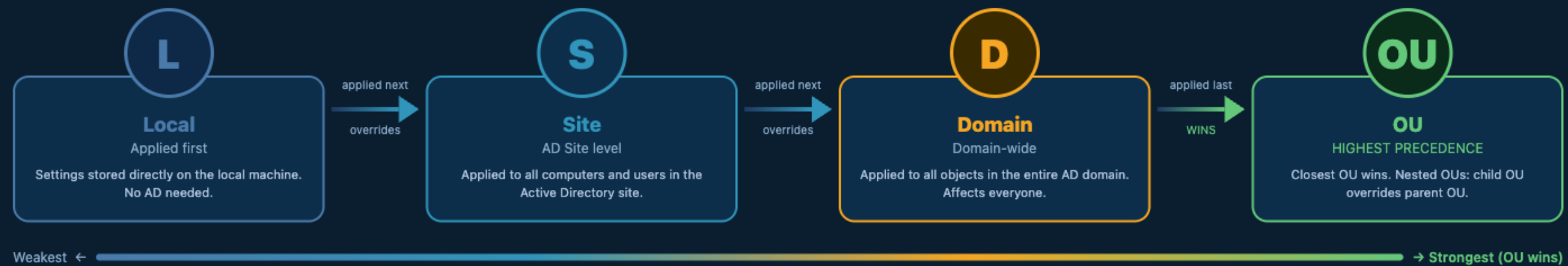
EXAM SHORTCUT — Rename 'Administrator' = attacker must guess username too, not just password

GPO Application Order — LSDOU

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GPO Application Order — LSDOU (Last Writer Wins)



EXAM TIP • If Local says "No screensaver" but OU says "Screensaver = company logo" → **OU wins**.
Conflicts are resolved by **override order**, not merge. The last policy applied on a conflicting setting is what takes effect.

Practice Question 3.1

D3 — Operating Systems



D3 — Operating Systems

QUESTION 3.1

A Windows domain administrator applies a password complexity policy via a GPO at the Domain level. A specific department OU also has a GPO that sets a different, less strict password policy. When a user in that OU logs on, which policy is enforced and why?

- A. The Domain-level GPO is enforced because domains have authority over all OUs.
- B. Both GPOs are merged and the stricter setting from each is applied.
- C. The OU-level GPO is enforced because OU policies are processed last in the LSDOU sequence and override Domain policies.
- D. The Local GPO is enforced because it is applied first and cannot be overridden by domain objects.

Answer — Question 3.1

D3 — Operating Systems



D3 — Operating Systems

ANSWER C

C. The OU-level GPO is enforced — OU policies are processed last and override Domain policies.

Why this is correct

GPOs are applied in LSDOU order: Local, Site, Domain, then Organizational Unit. Each subsequent layer can override the previous one. Since the OU is processed last, its GPO takes precedence over the Domain-level policy for users and computers in that OU.

Why the other options are wrong

- A — Domain GPOs do not automatically override OU GPOs. OU is processed after Domain and wins in conflicts.
- B — GPOs do not merge on a per-setting basis by strictness. The last-applied GPO wins the conflict.
- D — Local GPO is processed first and has the lowest priority; any domain-level policy will override it.

Linux File Permissions — chmod 750

Linux File Permissions — chmod 750

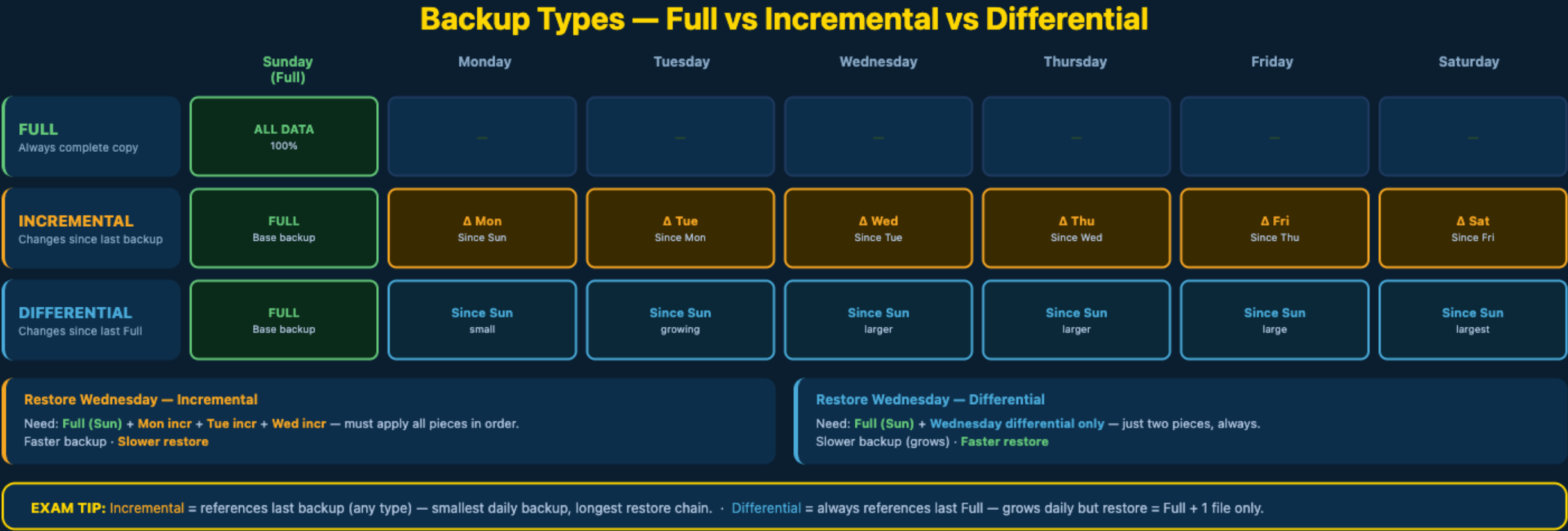


```
chmod 750 filename
```

Scope	Octal	Symbolic	What it means
Owner (User)	7	rwX	Read + Write + Execute — full access
Group	5	r-X	Read + Execute — can run, cannot modify
Others (World)	0	---	No permissions whatsoever — completely locked out

Common permissions: 755 = owner full, group+others r-x (public executables) · 644 = owner rw, group+others r (public files) · 600 = owner rw only (private keys)

Backup Types — Full vs Incremental vs Differential



Linux — Permissions, Signals, Files

D3 — Operating Systems



CHMOD — OCTAL PERMISSIONS

- Format: chmod [user][group][others] where each digit = sum of r=4, w=2, x=1
- 7 = rwx | 6 = rw- | 5 = r-x | 4 = r-- | 0 = ---
- chmod 640 → Owner: rw- (6) | Group: r-- (4) | Others: --- (0)
- chmod 755 → Owner: rwx (7) | Group: r-x (5) | Others: r-x (5)

EXAM SHORTCUT — chmod 640: owner=rw, group=r, others=none. 4=r, 2=w, 1=x, add them up

PROCESS SIGNALS

- SIGTERM (15) = polite request to stop — process CAN catch it and clean up first
- SIGKILL (9) = immediate kernel termination — CANNOT be caught or ignored
- Best practice: send SIGTERM first; escalate to SIGKILL only if process does not stop
- Only SIGKILL guarantees termination; SIGTERM may be ignored by a misbehaving process

EXAM SHORTCUT — SIGTERM = ask nicely (ignorable). SIGKILL = kernel forces it (cannot be ignored)

/etc/passwd vs /etc/shadow

- /etc/passwd — readable by ALL users; stores username, UID, GID, home dir, shell
- /etc/passwd — does NOT store passwords (shows "x" as placeholder)
- /etc/shadow — readable by ROOT only; stores hashed passwords + expiry/aging rules
- Split design: prevents unprivileged users from reading and cracking password hashes

EXAM SHORTCUT — /etc/passwd = public info only. /etc/shadow = root-only hashed passwords

Practice Question 3.2

D3 — Operating Systems



D3 — Operating Systems

QUESTION 3.2

A Linux administrator executes the command `chmod 750 deploy.sh` on a deployment script. Which permission set is correctly assigned to the file?

A. Owner: read/write/execute | Group: read/execute | Others: no access

B. Owner: read/write | Group: read/write/execute | Others: read only

C. Owner: read/write/execute | Group: read only | Others: read/execute

D. Owner: read only | Group: read/write | Others: no access

Answer — Question 3.2

D3 — Operating Systems



D3 — Operating Systems

ANSWER A

A. Owner: rwx (7) | Group: r-x (5) | Others: --- (0)

Why this is correct

7 = 4+2+1 = read+write+execute. 5 = 4+0+1 = read+execute. 0 = no permissions. chmod 750 gives the owner full access, the group read and execute, and blocks others entirely.

Why the other options are wrong

- B — 750 gives group r-x (5), not rwx (7). Group write would require digit 7 or 6 in the group position.
- C — Others position is 0 (no access), not 5 (r-x). Check the third octet.
- D — Owner digit 7 = rwx, not read only (4). Read only in the owner position would require chmod 4xx.

Backup + DR + Storage

D3 — Operating Systems



RPO vs RTO

- RPO (Recovery Point Objective) = max acceptable DATA LOSS measured in time
- Example: RPO = 4 hours means you can afford to lose up to 4 hours of data
- RTO (Recovery Time Objective) = max acceptable DOWNTIME to restore operations
- Example: RTO = 2 hours means systems must be back online within 2 hours

EXAM SHORTCUT — RPO = how old can the backup be? RTO = how fast must you recover?

BACKUP TYPES — SPEED vs RESTORE COMPLEXITY

- Full → backs up everything. Slow to run. Fastest to restore (1 set only)
- Differential → backs up all changes since the LAST FULL. Medium speed.
- Restore = Full + latest Differential only (2 sets)
- Incremental → backs up changes since the LAST BACKUP of any type. Fastest to run.
- Restore = Full + every Incremental in sequence (many sets)

EXAM SHORTCUT — Incremental = fastest backup, slowest restore. Differential = middle ground

RAID 5

- Minimum 3 disks required
- Distributed parity: parity data is spread across all disks, not a single dedicated disk
- Overhead = equivalent of exactly 1 disk (no matter how many disks in the array)
- Usable capacity = $(N - 1) \times \text{disk size}$. Survives 1 disk failure.
- RAID 0 = no parity (speed). RAID 1 = mirroring (50% overhead). RAID 5 = 1-disk overhead.

EXAM SHORTCUT — RAID 5: min 3 disks, 1-disk parity overhead, survives 1 failure

Practice Question 3.3

D3 — Operating Systems



D3 — Operating Systems

QUESTION 3.3

A system runs a full backup every Sunday. For Monday through Saturday it uses daily backups. A failure occurs on Friday morning. Which statement correctly compares the restoration process for incremental versus differential daily backups?

- A. Both strategies require restoring only the most recent daily backup file.
- B. Differential restore requires the Sunday full backup plus every daily differential from Monday to Thursday. Incremental requires only the Thursday file.
- C. Incremental restore requires the Sunday full backup plus every daily incremental from Monday through Thursday. Differential requires only the Sunday full backup plus the Thursday differential.
- D. RAID 5 parity automatically recovers data without requiring any backup restoration.

Answer — Question 3.3

D3 — Operating Systems



D3 — Operating Systems

ANSWER C

C. Incremental: Sunday full + every daily incremental (Mon-Thu). Differential: Sunday full + Thursday differential only.

Why this is correct

Incremental backups each capture only the changes since the previous backup of any kind. To restore, you need the full backup plus every subsequent incremental in order — miss one and the chain breaks. Differential backups capture all changes since the last full. Restore only needs the full backup plus the most recent differential — simpler and faster.

Why the other options are wrong

- A — Neither strategy can restore from a single file after a full failure; you always need at least the Sunday full backup as the base.
- B — This reverses the two strategies. It is the Incremental that requires every daily file; Differential needs only the latest one.
- D — RAID 5 protects against disk hardware failure, not against accidental deletion, corruption, or ransomware. Backups are still required.

Active Directory — FSMO Roles

D3 — Operating Systems



WHAT ARE FSMO ROLES?

- Flexible Single Master Operations — 5 special AD roles each held by exactly ONE DC
- Prevent conflicts for operations that cannot be done simultaneously on multiple DCs
- Forest-wide roles (one per forest): Schema Master, Domain Naming Master
- Domain-wide roles (one per domain): RID Master, PDC Emulator, Infrastructure Master

EXAM SHORTCUT — FSMO = one DC owns each special role. Can be transferred; cannot be shared.

THE 5 FSMO ROLES — KNOW EACH ONE

- SCHEMA MASTER — controls all changes to the AD schema (object class definitions)
- DOMAIN NAMING MASTER — manages adding and removing domains from the forest
- RID MASTER — allocates pools of Relative IDs (RIDs) to each domain controller
 - → DC builds unique SID = Domain SID + RID from its pool
 - → Every user, group, computer gets a unique SID this way
- PDC EMULATOR — handles time synchronisation, legacy auth, password change replication
- INFRASTRUCTURE MASTER — maintains cross-domain object references and phantoms

EXAM SHORTCUT — RID Master = allocates RID pools → DCs build unique SIDs. PDC Emulator = time sync + passwords.

EXAM FOCUS — RID MASTER vs DISTRACTORS

- Q: "Which role ensures every security principal gets a unique SID?" → RID MASTER
- Schema Master creates/modifies object TYPES, not individual object identities
- PDC Emulator handles TIME and PASSWORD synchronisation — not SID allocation
- Infrastructure Master handles cross-domain references — not RID pools
- Memory hook: RID = Relative ID → uniqueness → SID building

EXAM SHORTCUT — SID = Domain SID + RID. RID pools come from RID Master. Unique identities = RID Master.

Practice Question 3.4

D3 — Operating Systems



D3 — Operating Systems

QUESTION 3.4

A junior administrator needs to create 500 new user accounts across a domain. The domain controller servicing the requests needs a fresh allocation of identifiers before it can generate unique Security Identifiers (SIDs) for the new accounts. Which FSMO role is responsible for providing this allocation?

- A. PDC Emulator — synchronises time across domain controllers and handles legacy authentication requests.
- B. Schema Master — manages modifications to the Active Directory schema object class definitions.
- C. RID Master — allocates pools of Relative IDs to domain controllers, which combine them with the domain SID to build unique SIDs for new security principals.
- D. Infrastructure Master — maintains cross-domain group membership references and phantom objects.

Answer — Question 3.4

D3 — Operating Systems



D3 — Operating Systems

ANSWER C

C. RID Master — allocates RID pools to DCs so they can construct unique SIDs.

Why this is correct

Every Active Directory security principal (user, group, computer) gets a SID built from: Domain SID + Relative ID (RID). The RID Master is the single DC that allocates pools of RIDs to other DCs. When a DC runs low on RIDs it requests a new pool from the RID Master. Without this role, DCs could assign duplicate RIDs and create non-unique SIDs.

Why the other options are wrong

- A — The PDC Emulator handles time synchronisation (NTP hierarchy within the domain) and acts as the authoritative server for password changes and legacy NTLM auth. It plays no role in SID or RID allocation.
- B — The Schema Master controls changes to the schema: adding new object classes or attributes. It governs the structure of the directory, not the identity of individual objects.
- D — The Infrastructure Master maintains references to objects in other domains (cross-domain group memberships, phantom records). It does not allocate RIDs or SIDs.